

A stopping rule for regression trees

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Abstract

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1 Introduction

2 An iterative stopping test statistic

Consider iid trainingsdata $(X, Y), (X_i, Y_i)_{i=1}^N$ taking values in $\mathbb{R}^d \times \mathbb{R}$ and satisfying

$$\mathcal{L}(Y \mid X) = \mathcal{N}(\mu(X), \sigma^2)$$

for an unknown regression function $\mu: \mathbb{R}^d \rightarrow \mathbb{R}$ and unknown $\sigma^2 > 0$. Let (X^*, Y^*) be iid with respect to (X, Y) .

2.1 Motivation

Suppose we have a disjoint partition $\bigcup_{j=1}^J D_j = \mathbb{R}^d$ and a tree regressor

$$\hat{\mu}_0(x) = \hat{\mu}_0(x, (X_i, Y_i)_{i=1}^N) = \sum_{j=1}^J I\{x \in D_j\} \hat{m}_j, \quad (1)$$

where $\hat{m}_j = \frac{1}{n_{D_j}} \sum_{i=1}^N Y_i I\{X_i \in D_j\}$ with $n_{D_j} = |\{i: X_i \in D_j\}|$. Let

$$\mu_{D_j} = \mathbb{E}[Y \mid X \in D_j], \quad \sigma_{D_j}^2 = \mathbb{E}[(Y - \mu_{D_j})^2 \mid X \in D_j] = \text{Var}(Y \mid X \in D_j).$$

Note that $\hat{\mu}_0(x)$ is a random variable since it is a function of the trainingsdata. Denote by

$$\mathbf{MSEP}_j^0 = \mathbb{E} \left[(Y^* - \hat{\mu}_0(X^*))^2 \mid X^* \in D_j, n_{D_j} \right]$$

the generalization error of the predictor $\hat{\mu}_0$, given region D_j and given there are n_{D_j} trainings-datapoints in region D_j .

Lemma 1. *Then,*

$$\mathbf{MSEP}_j^0 = \sigma_{D_j}^2 \left(1 + \frac{1}{n_{D_j}} \right).$$

Fix a region j and let $A_j \dot{\cup} B_j = D_j$ be a split at the terminal node j . Let $\hat{\mu}_1$ be defined as in (1), but with $J + 1$ regions after splitting node j . We let

$$\mathbf{MSEP}_j^1 = \mathbb{E} \left[(Y^* - \hat{\mu}_1(X^*))^2 \mid X^* \in D_j, n_{A_j}, n_{B_j} \right]$$

be the generalization error of $\hat{\mu}_1$, given D_j and given there are n_{A_j} and n_{B_j} trainings-datapoints in A_j and B_j , respectively.

Lemma 2. *Then,*

$$\mathbf{MSEP}_j^1 = \sigma_{A_j}^2 \left(1 + \frac{1}{n_{A_j}} \right) \pi_{A_j|D_j} + \sigma_{B_j}^2 \left(1 + \frac{1}{n_{B_j}} \right) \pi_{B_j|D_j},$$

where $\pi_{A_j|D_j} = \mathbb{P}(X \in A_j \mid X \in D_j)$ and analogous for $\pi_{B_j|D_j}$.

The split at node j increased the mean squared error of prediction (generalisation error) in region D_j , if

$$\mathbf{MSEP}_j^0 < \mathbf{MSEP}_j^1. \quad (2)$$

Estimating the unknown parameters $\sigma_{A_j}^2, \pi_{A_j|D_j}$ by the unbiased estimators

$$\hat{\sigma}_{A_j}^2 = \frac{1}{n_{A_j} - 1} \sum_{i: X_i \in A_j} (Y_i - \bar{Y}_{A_j})^2, \quad \hat{\pi}_{A_j|D_j} = n_{A_j}/n_{D_j}, \quad (3)$$

where $\bar{Y}_{A_j} = 1/n_{A_j} \sum_{i: X_i \in A_j} Y_i$, and similarly for $\sigma_{B_j}^2, \pi_{B_j|D_j}$, this motivates the stopping test statistic

$$R_j = \frac{\widehat{\mathbf{MSEP}}_j^1}{\widehat{\mathbf{MSEP}}_j^0} = \frac{(n_{A_j} + 1)\hat{\sigma}_{A_j}^2 + (n_{B_j} + 1)\hat{\sigma}_{B_j}^2}{(n_{D_j} + 1)\hat{\sigma}_{D_j}^2} \quad (4)$$

for a split at node j . Following (2), the interpretation of $R_j \gg 1$ is that, with a high probability, the local generalization error has increased after splitting.

2.2 An alternative motivation: Covariance penalties

Assume fixed covariate trainings data $x_1, \dots, x_N$ and independent trainings-response-data $Y_1, \dots, Y_N$ satisfying

$$\mathcal{L}(Y_i) = \mathcal{L}(Y \mid X = x_i) = \mathcal{N}(\mu(x_i), \sigma^2)$$

for an unknown regression function $\mu: \mathbb{R}^d \rightarrow \mathbb{R}$ and unknown $\sigma^2 > 0$. Let $Y_1^*, \dots, Y_N^*$ be independent, independent of $(Y_i)_{i=1}^N$ and satisfying $\mathcal{L}(Y_i^*) = \mathcal{L}(Y_i)$ for $i = 1, \dots, N$. Define the in-sample mean squared error of prediction before and after a split at node j by

$$\mathbf{MSEP}_{\text{in},j}^0 = \frac{1}{n_{D_j}} \sum_{i: x_i \in D_j} (Y_i^* - \hat{\mu}_0(x_i))^2, \quad \mathbf{MSEP}_{\text{in},j}^1 = \frac{1}{n_{D_j}} \sum_{i: x_i \in D_j} (Y_i^* - \hat{\mu}_1(x_i))^2.$$

Lemma 3. *Then,*

$$\begin{aligned} \mathbb{E}[\mathbf{MSEP}_{\text{in},j}^0] &= \sigma_{D_j}^2 \left(1 + \frac{1}{n_{D_j}} \right), \\ \mathbb{E}[\mathbf{MSEP}_{\text{in},j}^1] &= \sigma_{A_j}^2 \left(1 + \frac{1}{n_{A_j}} \right) \pi_{A_j|D_j} + \sigma_{B_j}^2 \left(1 + \frac{1}{n_{B_j}} \right) \pi_{B_j|D_j}, \end{aligned}$$

where $\pi_{A_j|D_j} = \mathbb{P}(X \in A_j \mid X \in D_j)$ and analogous for $\pi_{B_j|D_j}$.

Choosing the same estimators as in (3), this approach hence motivates the same stopping testing statistic R_j as in (4).

2.3 Signal/No-signal hypothesis testing

Ideally, one should stop splitting once the generalization error increases for the first time, i.e. when condition (2) is satisfied. However, since the true values $\sigma_{D_j}^2, \pi_{A_j|D_j}$ and $\pi_{B_j|D_j}$ are unknown, we have to use the estimators in (3), and the resulting criterion $R_j > 1$ is subject to randomness in the estimators $\hat{\sigma}_{D_j}^2$ and $\hat{\pi}_{A_j|D_j}, \hat{\pi}_{B_j|D_j}$. For this reason, our approach is to carry out a sequence of hypotheses tests. More formally, for each potential splitting node, we let

$$H_0^j: \mu_{A_j} = \mu_{B_j} = \mu_{D_j}. \quad (5)$$

The interpretation of H_0^j is that the true regression function is flat on D_j , meaning that there is no signal left on D_j and no further split should be made. Denote by $F_{R_j}^0$ the distribution function of the stopping test statistic R_j , given H_0^j holds true (cf. section 2.4). Then, letting r_j be a true realization of R_j , $F_{R_j}^0(r_j)$ gives us a p -value for H_0^j .

2.4 The null distribution

For the sake of readability, we keep node j fixed and drop it from the notation in the following. That is, we write H_0, R, n, n_A, n_B , instead of $H_0^j, R_j, n_{D_j}, n_{A_j}, n_{B_j}$. Let

$$F_R^0(r) = \mathbb{P}_0(R \leq r)$$

be the distribution function of the stopping test statistic R from (4), given H_0 holds true.

Lemma 4 (Null distribution). *It holds that*

$$F_R^0(r) = \mathbb{P}_0(Z_r^1 \leq 0),$$

where, under H_0 ,

$$Z_r^1 \stackrel{d}{=} \lambda_A^r \chi_{n_A-1}^2 + \lambda_B^r \tilde{\chi}_{n_B-1}^2 + \lambda_{neg}^r \bar{\chi}_1^2, \quad (6)$$

with

$$\lambda_A^r = \frac{n_A + 1}{n_A - 1} - r \frac{n + 1}{n - 1}, \quad \lambda_B^r = \frac{n_B + 1}{n_B - 1} - r \frac{n + 1}{n - 1}, \quad \lambda_{neg}^r = -r \frac{n + 1}{n - 1}$$

and $\chi_{n_A-1}^2$, $\tilde{\chi}_{n_B-1}^2$ and $\bar{\chi}_1^2$ are independent chi-squared distributed random variables with $n_A - 1$, $n_B - 1$ and 1 degrees of freedom, respectively.

Using the Gil-Pelaez formula, we can compute $F_R^0(r)$ efficiently via

$$\mathbb{P}_0(Z_r^1 \leq 0) = \frac{1}{2} - \frac{1}{\pi} \int_{\mathbb{R}^+} \frac{\text{Im}(\phi_{Z_r^1}(t))}{t} dt,$$

where

$$\phi_{Z_r^1}(t) = (1 - 2\lambda_A^r it)^{-\frac{n_A-1}{2}} (1 - 2\lambda_B^r it)^{-\frac{n_B-1}{2}} (1 - 2\lambda_{neg}^r it)^{-\frac{1}{2}}$$

is the characteristic function of Z_r^1 under H_0 .

Corollary 1 (Distribution under the alternative). *We have*

$$F_R^1(r) = \mathbb{P}_1(Z_r^1 \leq 0),$$

where, under H_1 ,

$$Z_r^1 \stackrel{d}{=} \lambda_A^r \chi_{n_A-1}^2 + \lambda_B^r \tilde{\chi}_{n_B-1}^2 + \lambda_{neg}^r \bar{\chi}_1^2(\xi_{neg}), \quad (7)$$

with non-centrality parameter $\xi_{neg} = \frac{n_A n_B (\mu_A - \mu_B)^2}{n \sigma^2}$ and all other variables are defined as in Lemma 4.

In Figure 1, we set $(n_A, n_B) = (20, 15)$, $\sigma^2 = 1$ and compare the cdf under H_0 with three different versions of cdfs under H_1 .

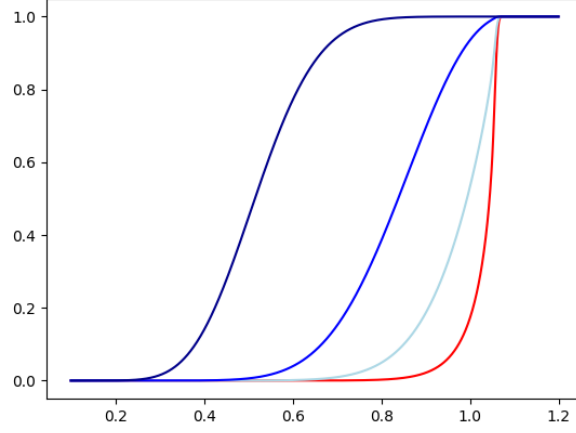


Figure 1: Comparison of $F_R^0(r)$ (red) and three different versions of $F_R^1(r)$ corresponding to $|\mu_A - \mu_B| = 0.5, 1, 2$ (lightblue, blue, darkblue).

The next result shows large sample size properties of R .

Lemma 5 (Large sample size). *Let $n \rightarrow \infty$ and assume that $n_A/n \rightarrow a \in (0, 1)$, $n_B/n \rightarrow 1 - a$. Then, under H_0 , R converges to 1 in probability.*

This is in line with the results in Figure 2.

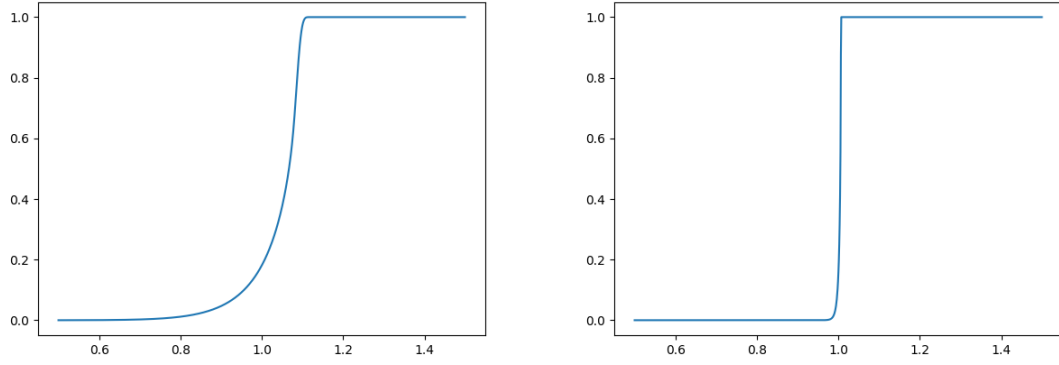


Figure 2: $F_R^0(r)$ for $(n_A, n_B) = (10, 13)$ and $(n_A, n_B) = (160, 170)$.

3 The stopping rule

4 Numerical illustration

5 Appendix: Proofs

Proof of Lemma 1. Since $\hat{\mu}_0$ is constant is on D_j , we have

$$\begin{aligned}
\mathbf{MSEP}_j^0 &= \mathbb{E}[(Y^* - \mu_{D_j})^2 \mid X^* \in D_j, n_{D_j}] + \mathbb{E}[(\mu_{D_j} - \hat{\mu}_0(X^*))^2 \mid X^* \in D_j, n_{D_j}] \\
&\quad + 2\mathbb{E}[(Y^* - \mu_{D_j})(\mu_{D_j} - \hat{\mu}_0(X^*)) \mid X^* \in D_j, n_{D_j}] \\
&= \sigma_{D_j}^2 + \mathbb{E}\left[\left(\left(\frac{1}{n_{D_j}} \sum_{i=1}^{n_{D_j}} Y_i\right) - \mu_{D_j}\right)^2 \mid n_{D_j}, X_1, \dots, X_{n_{D_j}} \in D_j\right] \\
&= \sigma_{D_j}^2 + \frac{1}{n_{D_j}^2} \mathbb{E}\left[\left(\sum_{i=1}^{n_{D_j}} (Y_i - \mu_{D_j})\right)^2 \mid n_{D_j}, X_1, \dots, X_{n_{D_j}} \in D_j\right] \\
&= \sigma_{D_j}^2 + \frac{1}{n_{D_j}^2} \sum_{i=1}^{n_{D_j}} \mathbb{E}[(Y_i - \mu_{D_j})^2 \mid X_i \in D_j] \\
&= \sigma_{D_j}^2 \left(1 + \frac{1}{n_{D_j}}\right).
\end{aligned}$$

In the second equality, the mixed term vanishes since, given $X^* \in D_j$, $\hat{\mu}_0(X^*, (X_i, Y_i)_{i=1}^N) = \hat{\mu}_0((X_i, Y_i)_{i=1}^N)$ only depends on trainings data and is hence independent of Y^* . $\square$

Proof of Lemma 2. Conditioning yields

$$\begin{aligned}
\mathbf{MSEP}_j^1 &= \mathbb{E}\left[(Y^* - \hat{\mu}_1(X^*))^2 \mid X^* \in A_j, n_{A_j}\right] \mathbb{P}(X^* \in A_j \mid X^* \in D_j) \\
&\quad + \mathbb{E}\left[(Y^* - \hat{\mu}_1(X^*))^2 \mid X^* \in B_j, n_{B_j}\right] \mathbb{P}(X^* \in B_j \mid X^* \in D_j) \\
&= \sigma_{A_j}^2 \left(1 + \frac{1}{n_{A_j}}\right) \pi_{A_j|D_j} + \sigma_{B_j}^2 \left(1 + \frac{1}{n_{B_j}}\right) \pi_{B_j|D_j},
\end{aligned}$$

where, in the last equality, we have applied Lemma 1 and the fact that $\hat{\mu}_1$ is constant on A_j and on B_j . $\square$

Proof of Lemma 4. Assume without loss of generality that $X_1, \dots, X_{n_A} \in A$ and $X_{n_A+1}, \dots, X_n \in B$. We wish to determine the distribution function F_R^0 of

$$R = \frac{\frac{n_A+1}{n_A-1} \sum_{i=1}^{n_A} (Y_i - \bar{Y}_A)^2 + \frac{n_B+1}{n_B-1} \sum_{i=n_A+1}^n (Y_i - \bar{Y}_B)^2}{\frac{n+1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2}$$

under H_0 , i.e. given that $\mu_A = \mu_B = \mu_D$. For this purpose, we note that for each $r > 0$,

$$F_R^0(r) = \mathbb{P}_0(R \leq r) = \mathbb{P}_0(Z_r \leq 0),$$

where

$$Z_r = \frac{n_A + 1}{n_A - 1} \sum_{i=1}^{n_A} (Y_i - \bar{Y}_A)^2 + \frac{n_B + 1}{n_B - 1} \sum_{i=n_A+1}^n (Y_i - \bar{Y}_B)^2 - r \frac{n+1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2. \quad (8)$$

Let

$$D_A = \left[\begin{array}{ccc|c} 1 & \cdots & 1 & \mathbf{0} \\ & & \vdots & \\ 1 & \cdots & 1 & \\ \hline & & \mathbf{0} & \mathbf{0} \end{array} \right], \quad I_A = \left[\begin{array}{c|c} \mathbf{I}_{n_A} & \mathbf{0} \\ \hline \mathbf{0} & \mathbf{0} \end{array} \right] \quad (9)$$

and D_B, I_B defined correspondingly, with the one-matrix and the identity matrix in the lower right block instead. We can then rewrite (8) as a quadratic form

$$Z_r = Y^T Q_r Y,$$

where

$$\begin{aligned} Q_r = & \frac{n_A + 1}{n_A - 1} \left(I_A - \frac{1}{n_A} D_A \right)^T \left(I_A - \frac{1}{n_A} D_A \right) + \\ & \frac{n_B + 1}{n_B - 1} \left(I_B - \frac{1}{n_B} D_B \right)^T \left(I_B - \frac{1}{n_B} D_B \right) - \\ & r \frac{n+1}{n-1} \left(I_n - \frac{1}{n} \mathbf{1} \right)^T \left(I_n - \frac{1}{n} \mathbf{1} \right). \end{aligned}$$

Here, $\mathbf{1}$ is the matrix of ones only. Using that all matrices in brackets are symmetric and idempotent,

$$\begin{aligned} Q_r = & \frac{n_A + 1}{n_A - 1} \left(I_A - \frac{1}{n_A} D_A \right) + \frac{n_B + 1}{n_B - 1} \left(I_B - \frac{1}{n_B} D_B \right) - r \frac{n+1}{n-1} \left(I_n - \frac{1}{n} \mathbf{1} \right) = \\ & \left[\begin{array}{ccc|ccc} \frac{n_A+1}{n_A} - \frac{r(n+1)}{n} & \frac{r(n+1)}{(n-1)n} - \frac{n_A+1}{(n_A-1)n_A} & \cdots & & & \\ \frac{r(n+1)}{(n-1)n} - \frac{n_A+1}{(n_A-1)n_A} & & & \frac{r(n+1)}{(n-1)n} & \cdots & \\ \vdots & & \ddots & \vdots & & \\ \hline & & & & & \\ & \frac{r(n+1)}{(n-1)n} & \cdots & \frac{n_B+1}{n_B} - \frac{r(n+1)}{n} & \frac{r(n+1)}{(n-1)n} - \frac{n_B+1}{(n_B-1)n_B} & \cdots \\ & \vdots & & \frac{r(n+1)}{(n-1)n} - \frac{n_B+1}{(n_B-1)n_B} & & \\ & & & \vdots & & \ddots \end{array} \right]. \end{aligned}$$

Q_r has four distinct eigenvalues

$$\begin{aligned}\lambda_A^r &= \frac{n_A + 1}{n_A - 1} - r \frac{n + 1}{n - 1}, \\ \lambda_B^r &= \frac{n_B + 1}{n_B - 1} - r \frac{n + 1}{n - 1}, \\ \lambda_{\text{neg}}^r &= -r \frac{n + 1}{n - 1}, \\ \lambda_0^r &= 0\end{aligned}$$

with corresponding eigenspaces

$$\begin{aligned}\text{Eig}_{\lambda_A} &= \text{span} \left(\begin{bmatrix} 1 \\ -1 \\ 0 \\ \vdots \\ 0 \\ \mathbf{0}_{n_B} \end{bmatrix}, \dots, \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ -1 \\ \mathbf{0}_{n_B} \end{bmatrix} \right), \quad \text{Eig}_{\lambda_B} = \text{span} \left(\begin{bmatrix} \mathbf{0}_{n_A} \\ 1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \dots, \begin{bmatrix} \mathbf{0}_{n_A} \\ 1 \\ 0 \\ \vdots \\ 0 \\ -1 \end{bmatrix} \right), \\ \text{Eig}_{\lambda_{\text{neg}}} &= \text{span} \left(\begin{bmatrix} n_B \\ \vdots \\ n_B \\ -n_A \\ \vdots \\ -n_A \end{bmatrix} \right), \quad \text{Eig}_{\lambda_0} = \text{span} \left(\begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \right)\end{aligned} \tag{10}$$

and hence geometric multiplicities $n_A - 1, n_B - 1, 1$ and 1 , respectively. Let

$$P = \left[\begin{array}{cc|cc} P_{11} & \mathbf{0} & n_B / \sqrt{n_A^2 n_B + n_A n_B^2} & 1 / \sqrt{n} \\ & & \vdots & \vdots \\ & & n_B / \sqrt{n_A^2 n_B + n_A n_B^2} & 1 / \sqrt{n} \\ \hline \mathbf{0} & P_{22} & -n_A / \sqrt{n_A^2 n_B + n_A n_B^2} & 1 / \sqrt{n} \\ & & \vdots & \vdots \\ & & -n_A / \sqrt{n_A^2 n_B + n_A n_B^2} & 1 / \sqrt{n} \end{array} \right],$$

where P_{11} (P_{22}) is an $n_A \times n_A - 1$ ($n_B \times n_B - 1$) matrix obtained by applying the Gram-Schmidt orthogonalization to the first n_A (last n_B) components of the vectors spanning Eig_{λ_A} (Eig_{λ_B}) from (10) above.

By Mathai, Provost, "Quadratic forms in random variables", page 29, formula (3.1a.4),

it follows that

$$Z_r = Y^T Q_r Y \stackrel{d}{=} \sigma^2 \left(\lambda_A^r \sum_{i=1}^{n_A-1} (U_i + b_i)^2 + \lambda_B^r \sum_{i=n_A}^{n-2} (U_i + b_i)^2 + \lambda_{\text{neg}}^r (U_{n-1} + b_{n-1})^2 \right). \quad (11)$$

Here, $U_1, \dots, U_{n-1}$ are iid standard normally distributed and, with

$$\bar{\mu}^T = (\bar{\mu}_A^T, \bar{\mu}_B^T) = (\mu_A, \dots, \mu_A, \mu_B, \dots, \mu_B) \in \mathbb{R}^{n_A} \times \mathbb{R}^{n_B},$$

$$b = \frac{1}{\sigma} P^T \bar{\mu} = \frac{1}{\sigma} \begin{bmatrix} P_{11}^T \bar{\mu}_A \\ P_{22}^T \bar{\mu}_B \\ \left(n_B \sum_{i=1}^{n_A} \bar{\mu}_i - n_A \sum_{i=n_A+1}^n \bar{\mu}_i \right) / \sqrt{n_A^2 n_B + n_A n_B^2} \\ \left(\sum_{i=1}^n \bar{\mu}_i \right) / \sqrt{n} \end{bmatrix}.$$

Non-central Chi-square Let $X_1, \dots, X_m$ independent normally distributed random variables with means $a_1, \dots, a_m$ and unit variances and set $\xi = \sum_{i=1}^m a_i^2$ (non-centrality parameter). Then, $\sum_{i=1}^m X_i^2$ follows a non-central chi-squared distribution with m degrees of freedom and is denoted by $\chi_m^2(\xi)$.

We would like to express (11) in terms of a linear combination of independent non-central chi-squares. For this purpose, we compute

$$\xi_A = \sum_{i=1}^{n_A-1} b_i^2, \quad \xi_B = \sum_{i=n_A}^{n-2} b_i^2, \quad \text{and} \quad \xi_{\text{neg}} = b_{n-1}^2.$$

Denote by $\tilde{P}_{11}^T$ the orthogonal $n_A \times n_A$ matrix obtained by adding $(1/\sqrt{n_A}, \dots, 1/\sqrt{n_A}) \in \mathbb{R}^{n_A}$ as an additional row to P_{11}^T . We then get

$$\begin{aligned} \xi_A &= \sum_{i=1}^{n_A-1} b_i^2 = \frac{1}{\sigma^2} \left(\sum_{i=1}^{n_A} \left(\left(\tilde{P}_{11}^T \bar{\mu}_A \right)_i \right)^2 - |(1/\sqrt{n_A}, \dots, 1/\sqrt{n_A}) \bar{\mu}_A|^2 \right) \\ &= \frac{1}{\sigma^2} \left(\sum_{i=1}^{n_A} \bar{\mu}_i^2 - \frac{1}{n_A} \left(\sum_{i=1}^{n_A} \bar{\mu}_i \right)^2 \right) = \frac{1}{\sigma^2} (n_A \mu_A^2 - n_A \mu_A^2) = 0, \end{aligned}$$

where we have used the orthogonality of $\tilde{P}_{11}$ in the third equality. Similarly, one gets

$$\xi_B = \frac{1}{\sigma^2} (n_B \mu_B^2 - n_B \mu_B^2) = 0$$

and

$$\xi_{\text{neg}} = \frac{\left(n_B \sum_{i=1}^{n_A} \bar{\mu}_i - n_A \sum_{i=n_A+1}^n \bar{\mu}_i \right)^2}{\sigma^2 (n_A^2 n_B + n_A n_B^2)} = \frac{n_A n_B (\mu_A - \mu_B)^2}{n \sigma^2}.$$

Altogether, we can rewrite (11) to get

$$Z_r \stackrel{d}{=} \sigma^2 \left(\lambda_A^r \chi_{n_A-1}^2 + \lambda_B^r \tilde{\chi}_{n_B-1}^2 + \lambda_{\text{neg}}^r \bar{\chi}_1^2(\xi_{\text{neg}}) \right), \quad (12)$$

where $\chi_{n_A-1}^2, \tilde{\chi}_{n_B-1}^2$ and $\bar{\chi}_1^2(\xi_{\text{neg}})$ are independent.

Under H_0 , we have $\mu_A = \mu_B$, which implies that $\xi_{\text{neg}} = 0$. It then follows that

$$Z_r \stackrel{d}{=} \sigma^2 \left(\lambda_A^r \chi_{n_A-1}^2 + \lambda_B^r \tilde{\chi}_{n_B-1}^2 + \lambda_{\text{neg}}^r \bar{\chi}_1^2 \right), \quad (13)$$

where all chi-squares are central and independent. Denoting the random variable in brackets by Z_r^1 , this means that

$$F_R^0(r) = \mathbb{P}_0(Z_r \leq 0) = \mathbb{P}_0(Z_r^1 \leq 0).$$

□

Proof of Lemma 5. Let $n \rightarrow \infty$ and assume that $n_A/n \rightarrow a \in (0, 1)$, $n_B/n \rightarrow 1 - a$. We first want to investigate the weak limit of Z_r^1 under H_0 , i.e. the limit of

$$\tilde{\lambda}_A^r \frac{\chi_{n_A-1}^2}{n_A-1} + \tilde{\lambda}_B^r \frac{\tilde{\chi}_{n_B-1}^2}{n_B-1} + \lambda_{\text{neg}}^r \bar{\chi}_1^2, \quad (14)$$

where all chi-squares are independent and

$$\begin{aligned} \tilde{\lambda}_A^r &= \frac{(n_A+1)(n-1) - r(n+1)(n_A-1)}{n-1}, \\ \tilde{\lambda}_B^r &= \frac{(n_B+1)(n-1) - r(n+1)(n_B-1)}{n-1}, \quad \lambda_{\text{neg}}^r = -r \frac{n+1}{n-1}. \end{aligned}$$

The first two standardised chi-squares in (14) each converge to 1 in probability. Moreover,

$$\begin{aligned} \lambda_{\text{neg}}^r &\rightarrow -r, \\ \tilde{\lambda}_A^r &= \frac{2n_B}{n-1} + \frac{(1-r)(n+1)(n_A-1)}{n-1} \rightarrow \begin{cases} 2(1-a), & r = 1 \\ -\infty, & r > 1 \\ +\infty, & r < 1 \end{cases} \end{aligned}$$

and similarly

$$\tilde{\lambda}_B^r \rightarrow \begin{cases} 2a, & r = 1 \\ -\infty, & r > 1 \\ +\infty, & r < 1. \end{cases}$$

Hence, under H_0 , we get convergence in distribution by the Slutsky Theorem

$$Z_r^1 \rightarrow Z_r^{1,\infty} := \begin{cases} 2 - \bar{\chi}_1, & r = 1 \\ -\infty, & r > 1 \\ +\infty, & r < 1. \end{cases}$$

Since $Z_r^{1,\infty}$ has no mass at 0 for any $r > 0$, this implies that

$$F_R^0(r) = \mathbb{P}_0(Z_r^1 \leq 0) \rightarrow \mathbb{P}_0(Z_r^{1,\infty} \leq 0) = \begin{cases} \mathbb{P}(\bar{\chi}_1 \geq 2), & r = 1 \\ 1, & r > 1 \\ 0, & r < 1. \end{cases}$$

This shows that the ratio R converges to 1 under H_0 in distribution (and hence in probability). Note that the limit does not depend on a . □

References